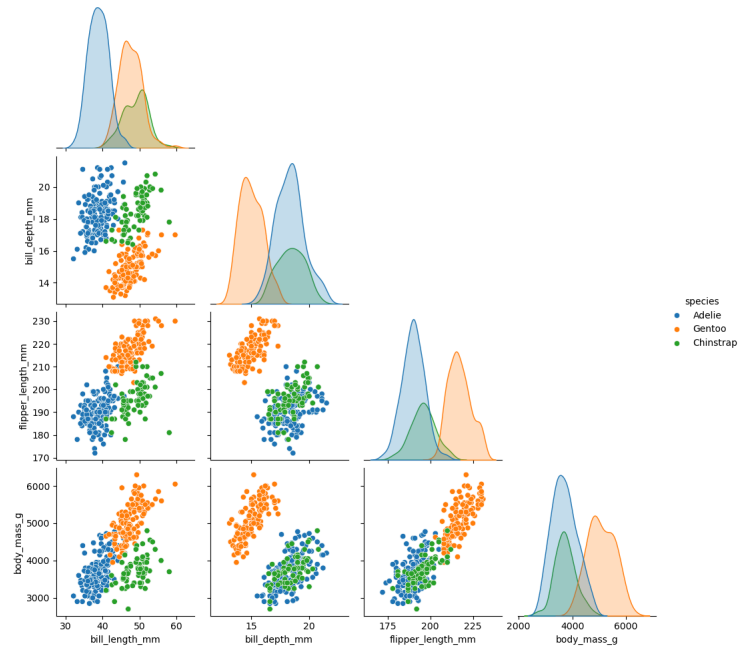


Foundations of Data Science

Lecture 4: Latent variable models

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What if we had not been given the species labels? Which unobserved quantities would explain what we see?

Probabilistic modeling of data

Recap from Lecture 2

A measurement is what a process produces from an entity $\omega \in \Omega$ (a penguin) under measurement conditions $\xi \in \Xi$ (the scale, the observer, the day),

$$f : \Omega \times \Xi \rightarrow \mathcal{X}.$$

Neither ω nor ξ is observed, only $\mathbf{x} = f(\omega, \xi)$.

Given a distribution $p(\omega, \xi)$ over entities and conditions, f induces the **data distribution**

$$p_r(\mathbf{x}) = \iint p(\omega, \xi) \delta(\mathbf{x} - f(\omega, \xi)) d\omega d\xi.$$

A **parametric probabilistic model** encodes assumptions about how data are generated. It is specified by a parametric family

$$\mathcal{P} = \{p(\mathbf{x} \mid \theta) : \theta \in \Theta\},$$

where $p(\mathbf{x} \mid \theta)$ is a probability distribution over $\mathcal{X}$, θ are parameters, and Θ is the parameter space.

$p(\mathbf{x} \mid \theta)$ is not the data distribution $p_r(\mathbf{x})$, only a model of it! A *true* parameter exists only if p_r belongs to the family $\mathcal{P}$.



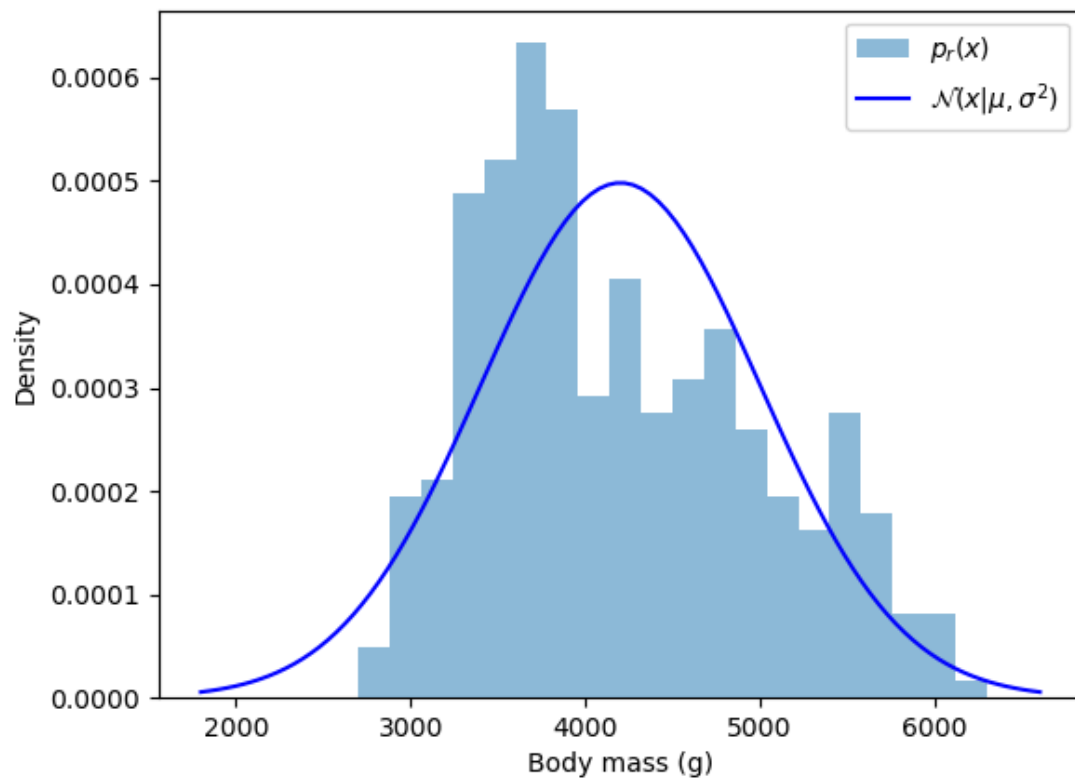
Example. Body masses of penguins could be modeled as

$$p(x \mid \mu, \sigma^2) = \mathcal{N}(x \mid \mu, \sigma^2),$$

where the parameters are $\theta = (\mu, \sigma^2)$, with $\mu \in \mathbb{R}$ and $\sigma^2 > 0$.

This would assume

- body masses cluster around a central value μ ,
- variability is symmetric and controlled by σ^2 ,
- extreme values are rare (body masses are normally distributed).



Frequentist inference

In the Frequentist framework, θ is treated as an unknown but fixed quantity to be estimated from observed data $\mathbf{x}_{\text{obs}}$. The data are assumed to be generated from the model for some unknown parameter value θ^* ,

$$\mathbf{x}_{\text{obs}} \sim p(\mathbf{x} \mid \theta^*).$$

Fitting the model to data consists in finding a point estimate $\hat{\theta}$ of θ^* (or a confidence region thereof) that best explains the observed data.

This θ^* lives inside the model: it is the parameter of the member of $\mathcal{P}$ the data are assumed to come from. If the family contains no such member, $\hat{\theta}$ estimates the parameter of the member closest to p_r .

Bayesian inference

In the Bayesian framework, θ is treated as a random variable with prior distribution $p(\theta)$ encoding beliefs about plausible parameter values before observing any data.

A **Bayesian model** therefore specifies a joint distribution over data and parameters,

$$p(\mathbf{x}, \theta) = p(\mathbf{x} \mid \theta)p(\theta),$$

where

- $p(\mathbf{x} \mid \theta)$ is the likelihood,
- $p(\theta)$ is the prior over parameters.

Fitting a Bayesian model to observed data $\mathbf{x}_{\text{obs}}$ consists in computing the posterior distribution of the parameters given the data. Using Bayes' rule,

$$p(\theta \mid \mathbf{x}_{\text{obs}}) = \frac{p(\mathbf{x}_{\text{obs}} \mid \theta)p(\theta)}{p(\mathbf{x}_{\text{obs}})}.$$

Depending on the structure of the model, this computation may be easy, difficult, or even intractable.

Point estimate, confidence interval, posterior

Three answers to the same question, for the mean body mass μ of the 342 penguins:

- A **point estimate**, $\hat{\mu} = 4202$ g, the value that maximizes the likelihood. One number, and nothing about its precision.
- A 95% **confidence interval**, $[4117, 4287]$ g. If the data were drawn again and again from the model, 95% of the intervals built this way would contain μ^* . This one either does or does not.
- A **posterior** $p(\mu \mid \mathbf{x}_{\text{obs}})$, here with a flat prior. Its 95% credible interval is the same $[4117, 4287]$ g, but it now says that, given the data and the prior, μ lies in it with probability 0.95.

The confidence interval is a statement about the procedure; the posterior is a statement about μ .

Prior predictive checks

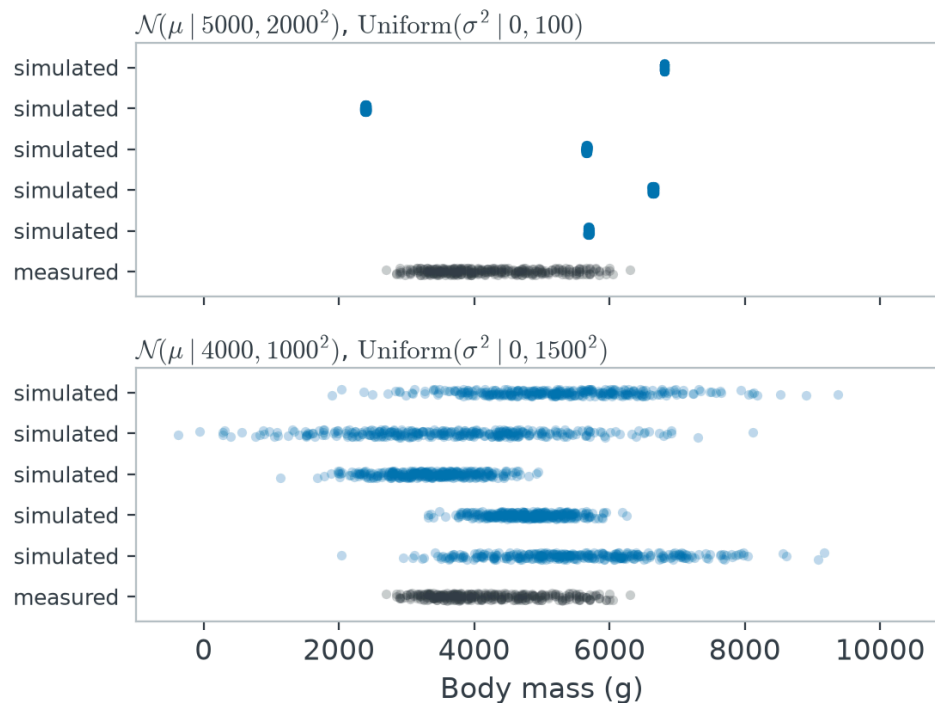
Often, the forward model $p(\mathbf{x} \mid \theta)$ is understood, but the prior $p(\theta)$ is more subjective and harder to justify.

The consequences of prior choices in the context of the generative model can be assessed through **prior predictive checks**, which involve simulating data from the model using only the prior distributions, without conditioning on any observed data.

The prior predictive distribution is given by

$$p(\mathbf{x}) = \int p(\mathbf{x} \mid \theta)p(\theta)d\theta.$$

This distribution defines the data that we expect to observe under the model assumptions encoded in the prior. It should be examined to ensure that it aligns with domain knowledge and expectations about the data.



Five colonies of 342 penguins simulated from the prior (top), and from a wider prior on σ^2 (bottom), against the penguins actually measured.

Latent variable models

Joint distribution

A **latent variable model** is a probabilistic model that assumes unobserved (latent) variables $\mathbf{z}$ that mediate the relationship between observed data $\mathbf{x}$ and model parameters θ .

It specifies a joint distribution over observed variables, latent variables, and parameters,

$$p(\mathbf{x}, \mathbf{z}, \theta) = p(\mathbf{x} \mid \mathbf{z}, \theta)p(\mathbf{z} \mid \theta)p(\theta),$$

where $\mathbf{x}$ is the observed data, $\mathbf{z}$ are the latent variables, and θ are the parameters.

More generally, for a dataset of N observations $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, a latent variable model specifies a joint distribution

$$p(\mathbf{x}_{1:N}, \mathbf{z}_{1:N}, \theta) = \left(\prod_{i=1}^N p(\mathbf{x}_i \mid \mathbf{z}_i, \theta) p(\mathbf{z}_i \mid \theta) \right) p(\theta),$$

where $\mathbf{z}_i$ are the latent variables associated with observation $\mathbf{x}_i$.

The factorization assumes that each observation $\mathbf{x}_i$ is mediated by its own latent variable $\mathbf{z}_i$, and that observations are conditionally independent given their latent variables and the parameters. All are governed by shared parameters θ .

Marginal likelihood

The latent variables are not observed. Integrating them out gives the likelihood of the data alone,

$$p(\mathbf{x} \mid \theta) = \int p(\mathbf{x} \mid \mathbf{z}, \theta) p(\mathbf{z} \mid \theta) d\mathbf{z},$$

or, for N conditionally independent observations,

$$p(\mathbf{x}_{1:N} \mid \theta) = \prod_{i=1}^N \int p(\mathbf{x}_i \mid \mathbf{z}_i, \theta) p(\mathbf{z}_i \mid \theta) d\mathbf{z}_i.$$

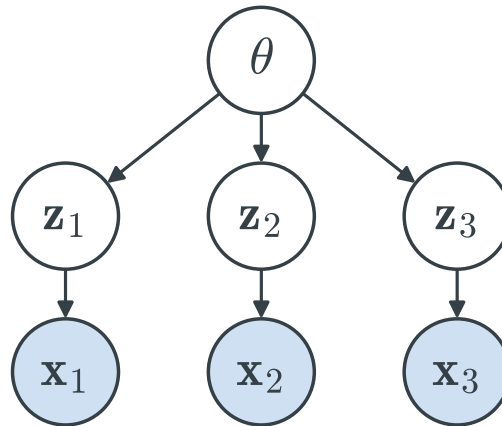
This integral is where the difficulty of latent variable models lies: it is what maximum likelihood maximizes, and it is the normalizer of the posterior over the latent variables,

$$p(\mathbf{z} \mid \mathbf{x}, \theta) = \frac{p(\mathbf{x} \mid \mathbf{z}, \theta) p(\mathbf{z} \mid \theta)}{p(\mathbf{x} \mid \theta)}.$$

Graphical model representation

Latent variable models can be represented using graphical models, where nodes represent variables (observed, latent, or parameters) and edges represent (possible) dependencies between them.

The graphical model illustrates the structure of the factorization of the joint distribution and the flow of the generative process.



$$p(\mathbf{x}_{1:3}, \mathbf{z}_{1:3}, \theta) = \left(\prod_{i=1}^3 p(\mathbf{x}_i \mid \mathbf{z}_i) p(\mathbf{z}_i \mid \theta) \right) p(\theta)$$

Shaded nodes represent observed variables, unshaded nodes represent latent variables or parameters.

Here $\mathbf{x}_i$ depends on θ only through $\mathbf{z}_i$. In general, θ may also point directly at $\mathbf{x}_i$.

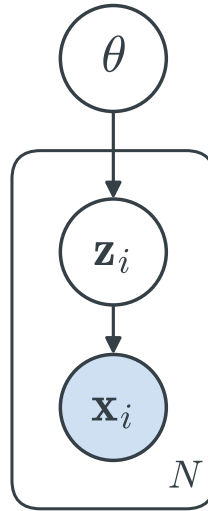


Plate notation can be used to compactly represent repeated structures in the graphical model.

z_i is **local**, θ is **global**.

Hyperparameters

Conditional distributions in a latent variable model may depend on additional parameters called **hyperparameters**, denoted $\alpha, \beta, \dots$. These are assumed to be fixed nonrandom quantities¹.

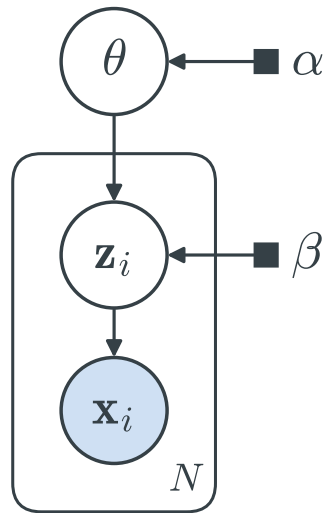
For instance, the prior distribution of parameters may depend on hyperparameters,

$$p(\theta \mid \alpha),$$

or the prior distribution of latent variables may depend on hyperparameters,

$$p(\mathbf{z} \mid \theta, \beta).$$

¹: Estimating hyperparameters from data is possible; this is empirical Bayes, discussed in Lecture 8.



Small squares denote fixed hyperparameters.

Inference

Fitting a latent variable model to observed data $\mathbf{x}_{\text{obs}}$ consists in computing the posterior distribution of latent variables and parameters given the data. Using Bayes' rule,

$$p(\mathbf{z}, \theta \mid \mathbf{x}_{\text{obs}}) = \frac{p(\mathbf{x}_{\text{obs}} \mid \mathbf{z}, \theta)p(\mathbf{z} \mid \theta)p(\theta)}{p(\mathbf{x}_{\text{obs}})}.$$

The posterior distribution is used to examine the particular hidden structure that is manifested in the observed data. It can also be used to make predictions about new, unseen data, through the posterior predictive distribution,

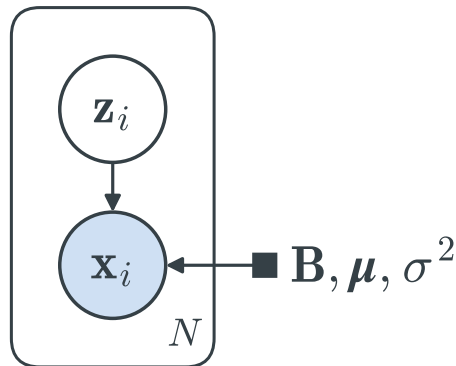
$$p(\mathbf{x}_{\text{new}} \mid \mathbf{x}_{\text{obs}}) = \iint p(\mathbf{x}_{\text{new}} \mid \mathbf{z}, \theta)p(\mathbf{z}, \theta \mid \mathbf{x}_{\text{obs}}) d\mathbf{z} d\theta.$$

Latent variables appear wherever the quantity that matters cannot be recorded:

- **Astronomy**: the distance of a star, behind a noisy parallax (Example 4).
- **Engineering**: the position and velocity of a vehicle, behind its sensors (state-space models, Lecture 5).
- **Education**: the ability of a student, behind right and wrong answers (item response theory).
- **Genetics**: the ancestral populations a genome is mixed from, behind its alleles (admixture models, the same structure as the topic models below).
- **Epidemiology**: how many people are actually infected, behind the cases a health system reports.

Example 1: (Probabilistic) PCA

In probabilistic PCA, each observation $\mathbf{x}_i \in \mathbb{R}^d$ is assumed to be generated from a lower-dimensional latent variable $\mathbf{z}_i \in \mathbb{R}^m$ through a linear transformation plus Gaussian noise.



The joint distribution $p(\mathbf{z}, \mathbf{x} \mid \mathbf{B}, \boldsymbol{\mu}, \sigma^2)$ factorizes as $p(\mathbf{z})p(\mathbf{x} \mid \mathbf{z}, \mathbf{B}, \boldsymbol{\mu}, \sigma^2)$, where

- $p(\mathbf{z}) = \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I})$ assumes latent variables are standard Gaussian,
- $p(\mathbf{x} \mid \mathbf{z}, \mathbf{B}, \boldsymbol{\mu}, \sigma^2) = \mathcal{N}(\mathbf{x} \mid \mathbf{B}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I})$ assumes a linear Gaussian observation model, with $\mathbf{B} \in \mathbb{R}^{d \times m}$ the loading matrix, $\boldsymbol{\mu} \in \mathbb{R}^d$ the mean vector, and σ^2 the noise variance.

Therefore, using Gaussian identities, the joint distribution is Gaussian and can be written as

$$p(\mathbf{z}, \mathbf{x} \mid \mathbf{B}, \boldsymbol{\mu}, \sigma^2) = \mathcal{N} \left(\begin{bmatrix} \mathbf{z} \\ \mathbf{x} \end{bmatrix} \mid \begin{bmatrix} \mathbf{0} \\ \boldsymbol{\mu} \end{bmatrix}, \begin{bmatrix} \mathbf{I} & \mathbf{B}^T \\ \mathbf{B} & \mathbf{B}\mathbf{B}^T + \sigma^2\mathbf{I} \end{bmatrix} \right).$$

The posterior distribution $p(\mathbf{z} \mid \mathbf{x}, \mathbf{B}, \boldsymbol{\mu}, \sigma^2)$ is also Gaussian,

$$p(\mathbf{z} \mid \mathbf{x}, \mathbf{B}, \boldsymbol{\mu}, \sigma^2) = \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_{z|x}, \boldsymbol{\Sigma}_{z|x}),$$

where

- $\boldsymbol{\mu}_{z|x} = \mathbf{B}^T(\mathbf{B}\mathbf{B}^T + \sigma^2\mathbf{I})^{-1}(\mathbf{x} - \boldsymbol{\mu})$ is the posterior mean,
- $\boldsymbol{\Sigma}_{z|x} = \mathbf{I} - \mathbf{B}^T(\mathbf{B}\mathbf{B}^T + \sigma^2\mathbf{I})^{-1}\mathbf{B}$ is the posterior covariance.

When $\sigma^2 \rightarrow 0$, and writing the posterior in the equivalent form $\boldsymbol{\mu}_{z|x} = (\mathbf{B}^T \mathbf{B} + \sigma^2 \mathbf{I})^{-1} \mathbf{B}^T (\mathbf{x} - \boldsymbol{\mu})$ and $\boldsymbol{\Sigma}_{z|x} = \sigma^2 (\mathbf{B}^T \mathbf{B} + \sigma^2 \mathbf{I})^{-1}$,

- $\boldsymbol{\mu}_{z|x} \rightarrow (\mathbf{B}^T \mathbf{B})^{-1} \mathbf{B}^T (\mathbf{x} - \boldsymbol{\mu})$. If the columns of $\mathbf{B}$ are orthonormal, this is $\mathbf{B}^T (\mathbf{x} - \boldsymbol{\mu})$, the PCA projection of $\mathbf{x}$ onto the subspace spanned by the columns of $\mathbf{B}$.
- $\boldsymbol{\Sigma}_{z|x} \rightarrow \mathbf{0}$, so the posterior collapses to a point mass at that projection.

Probabilistic PCA recovers classical PCA in the limit of vanishing noise!

The parameters $\theta = (\mathbf{B}, \boldsymbol{\mu}, \sigma^2)$ can be estimated from data $\mathbf{x}_{1:N}$ by maximum (marginal) likelihood,

$$(\hat{\mathbf{B}}, \hat{\boldsymbol{\mu}}, \hat{\sigma}^2) = \arg \max_{\mathbf{B}, \boldsymbol{\mu}, \sigma^2} \prod_{i=1}^N p(\mathbf{x}_i \mid \mathbf{B}, \boldsymbol{\mu}, \sigma^2),$$

where $p(\mathbf{x} \mid \mathbf{B}, \boldsymbol{\mu}, \sigma^2) = \int p(\mathbf{x} \mid \mathbf{z}, \mathbf{B}, \boldsymbol{\mu}, \sigma^2) p(\mathbf{z}) d\mathbf{z}$ is the marginal likelihood.

Since the joint distribution is Gaussian, the marginal likelihood is also Gaussian,

$$p(\mathbf{x} \mid \mathbf{B}, \boldsymbol{\mu}, \sigma^2) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \mathbf{B}\mathbf{B}^T + \sigma^2\mathbf{I}).$$

This is a point estimate of θ , not a posterior over it: the latent variables $\mathbf{z}_i$ are integrated out, the parameters are not.

Therefore, writing $\Sigma = \mathbf{B}\mathbf{B}^T + \sigma^2\mathbf{I}$, maximum likelihood estimation reduces to

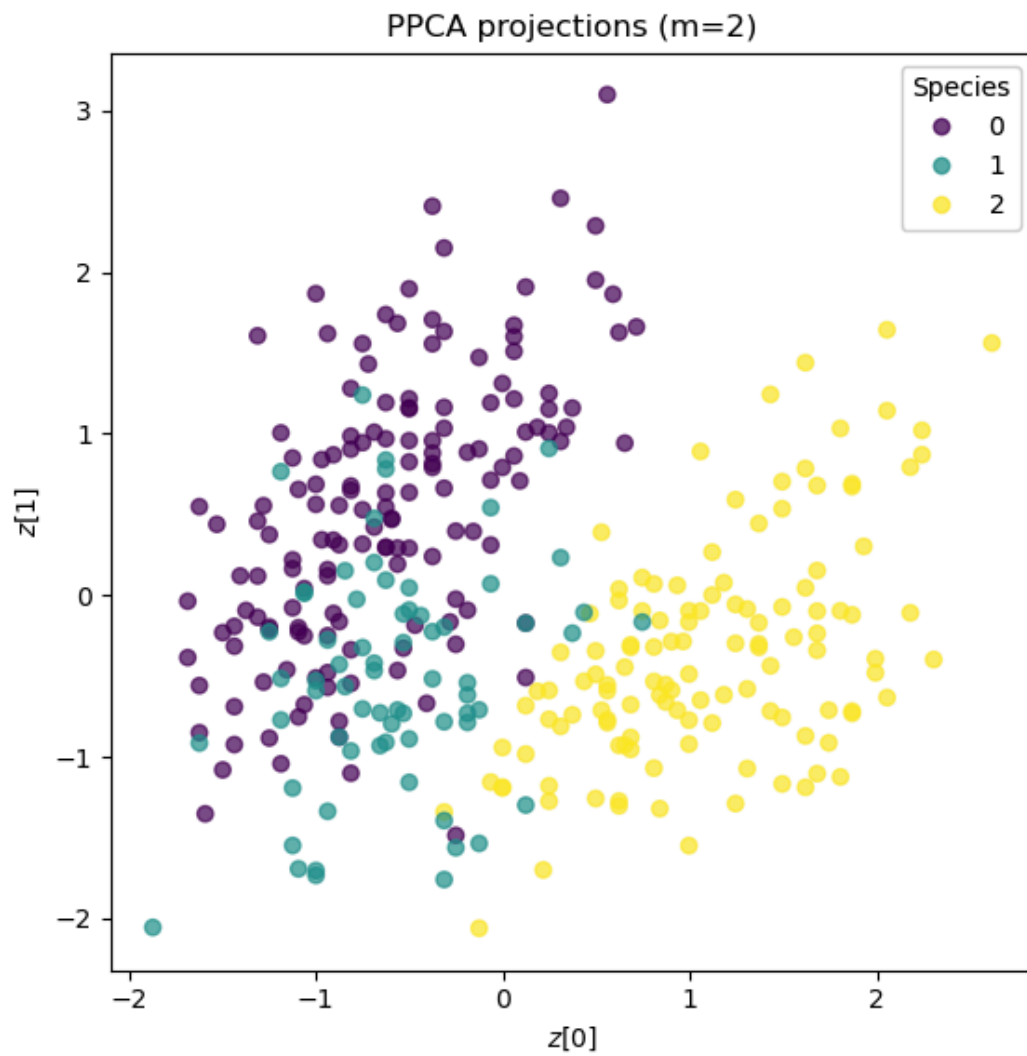
$$\begin{aligned}
 (\hat{\mathbf{B}}, \hat{\boldsymbol{\mu}}, \hat{\sigma}^2) &= \arg \max_{\mathbf{B}, \boldsymbol{\mu}, \sigma^2} \prod_{i=1}^N \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}, \Sigma) \\
 &= \arg \min_{\mathbf{B}, \boldsymbol{\mu}, \sigma^2} \sum_{i=1}^N (\mathbf{x}_i - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) + N \log |\Sigma| \\
 &= \arg \min_{\mathbf{B}, \boldsymbol{\mu}, \sigma^2} N \operatorname{tr}(\Sigma^{-1} \mathbf{S}) + N \log |\Sigma|,
 \end{aligned}$$

where $\mathbf{S} = \frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T$ is the sample covariance matrix.

The solution can be derived in closed form, yielding

- $\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i$ (the sample mean),
- $\hat{\mathbf{B}} = \mathbf{U}_m (\boldsymbol{\Lambda}_m - \hat{\sigma}^2 \mathbf{I})^{1/2} \mathbf{R}$, where $\mathbf{U}_m$ holds the top m eigenvectors of $\mathbf{S}$, $\boldsymbol{\Lambda}_m$ the corresponding eigenvalues, and $\mathbf{R}$ is an arbitrary rotation matrix,
- $\hat{\sigma}^2 = \frac{1}{d-m} \sum_{j=m+1}^d \lambda_j$, where λ_j are the eigenvalues of $\mathbf{S}$.

Since $\mathbf{R}$ is free, $\mathbf{B}$ is identified only up to a rotation: the model pins down the latent subspace, not the coordinates within it.





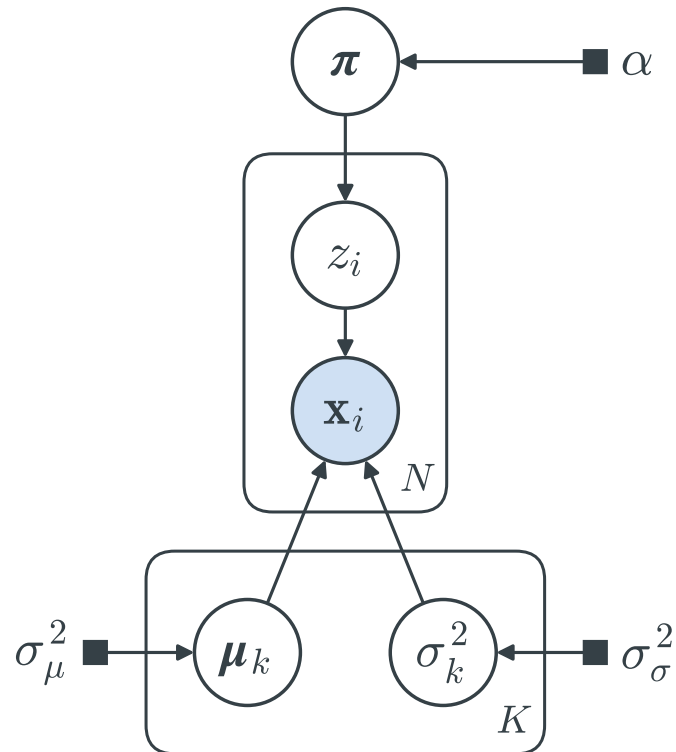
Deriving PCA from a latent variable model provides a **probabilistic interpretation of PCA projections as the most likely latent variables that could have generated the observed data.**

It also enables direct extensions such as

- Independent Component Analysis (ICA), which assumes non-Gaussian latent variables,
- Factor Analysis, which assumes a more general noise covariance structure,
- Bayesian PCA, which places a prior distribution over the hyperparameters.

Example 2: Mixture models

Mixture models assume that data are generated from a mixture of several underlying distributions, each corresponding to a different cluster or component.



For a Gaussian mixture model with K components, each observation $\mathbf{x}_i \in \mathbb{R}^d$ is assumed to be generated by first selecting a component $z_i \in \{1, \dots, K\}$ according to a categorical distribution, then sampling $\mathbf{x}_i$ from a Gaussian distribution associated with that component.

The parameters are $\theta = (\boldsymbol{\pi}, \boldsymbol{\mu}_{1:K}, \sigma_{1:K}^2)$, where $\boldsymbol{\pi}$ holds the mixture weights. The joint distribution $p(\theta, z_{1:N}, \mathbf{x}_{1:N} \mid \alpha, \sigma_\mu^2, \sigma_\sigma^2)$ factorizes as

$$p(\boldsymbol{\pi} \mid \alpha) \prod_{k=1}^K p(\boldsymbol{\mu}_k \mid \sigma_\mu^2) p(\sigma_k^2 \mid \sigma_\sigma^2) \prod_{i=1}^N p(z_i \mid \boldsymbol{\pi}) p(\mathbf{x}_i \mid z_i, \boldsymbol{\mu}_{z_i}, \sigma_{z_i}^2),$$

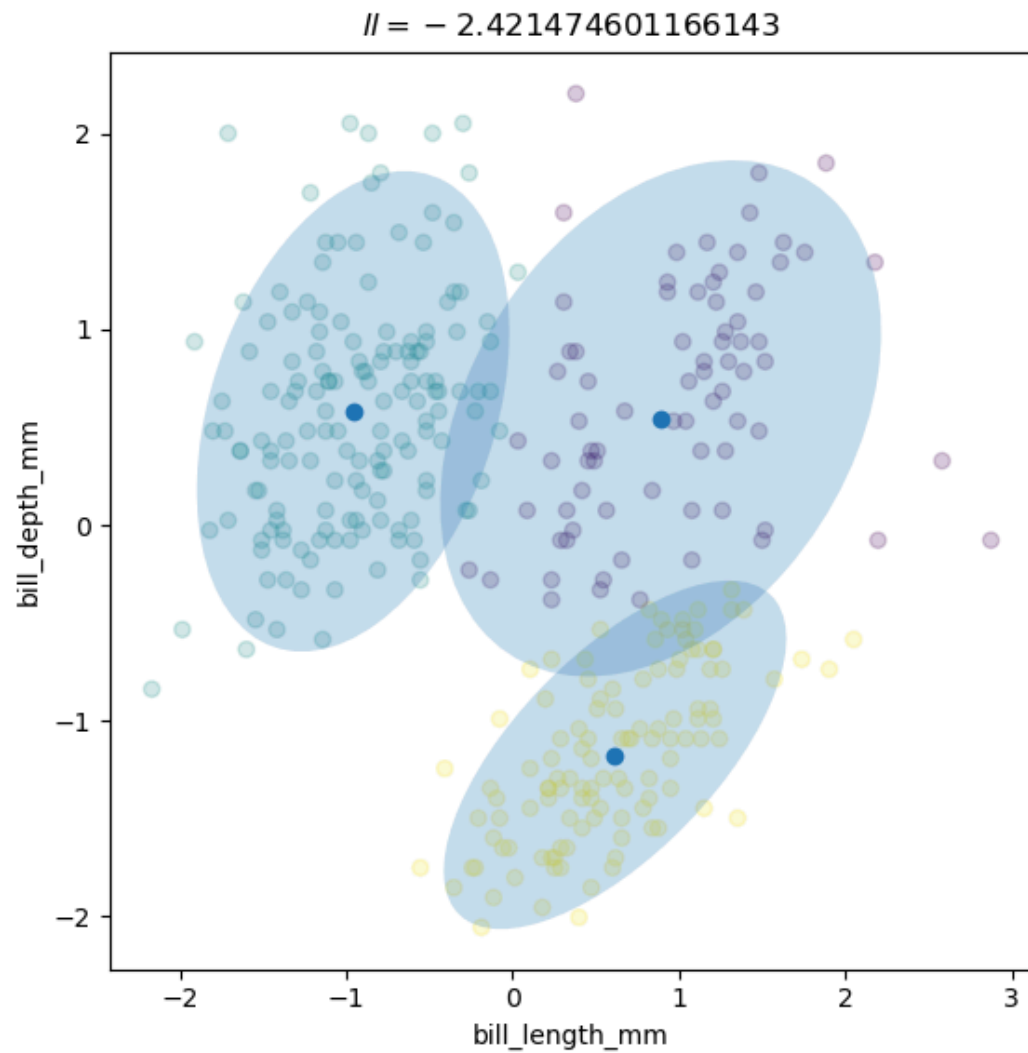
where

- $p(\boldsymbol{\pi} \mid \alpha) = \text{Dirichlet}(\boldsymbol{\pi} \mid \alpha)$ is the prior over mixture weights,
- $p(\boldsymbol{\mu}_k \mid \sigma_\mu^2) = \mathcal{N}(\boldsymbol{\mu}_k \mid \mathbf{0}, \sigma_\mu^2 \mathbf{I})$ is the prior over component means,
- $p(\sigma_k^2 \mid \sigma_\sigma^2) = \text{Lognormal}(\sigma_k^2 \mid 0, \sigma_\sigma^2)$ is the prior over component variances,
- $p(z_i \mid \boldsymbol{\pi}) = \text{Categorical}(z_i \mid \boldsymbol{\pi})$ is the distribution over components,
- $p(\mathbf{x}_i \mid z_i, \boldsymbol{\mu}_{z_i}, \sigma_{z_i}^2) = \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_{z_i}, \sigma_{z_i}^2 \mathbf{I})$ is the Gaussian observation model.

Computing the posterior distribution $p(\theta, z_{1:N} \mid \mathbf{x}_{1:N}, \alpha, \sigma_\mu^2, \sigma_\sigma^2)$ amounts to solving a clustering problem, where each component corresponds to a cluster and the latent variables z_i indicate cluster membership of each observation.

The posterior is typically intractable. Sampling from it is the subject of Lecture 6 (MCMC), approximating it by a simpler distribution that of Lecture 9 (variational inference); Lecture 8 (EM) settles instead for a point estimate of θ , with the z_i integrated out.

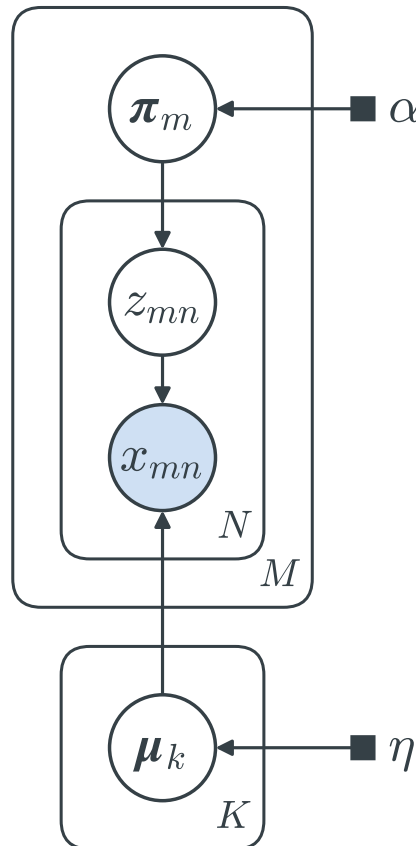
A mixture is also identified only up to a permutation of its components: relabelling them leaves the distribution unchanged, so the posterior has $K!$ equivalent modes.



Example 3: Mixed membership models

Nested sets of latent variables can also be used to model more complex generative structures.

For instance, in mixed membership models of text documents (**latent Dirichlet allocation**), each document is assumed to be generated from a mixture of topics, where each topic is characterized by a distribution over words.



Posterior inference in mixed membership models can be used to discover the underlying topics in a corpus of documents and to infer the topic proportions for each document.

gene 0.04
dna 0.02
genetic 0.01
...

life 0.02
evolve 0.01
organism 0.01
...

brain 0.04
neuron 0.02
nerve 0.01
...

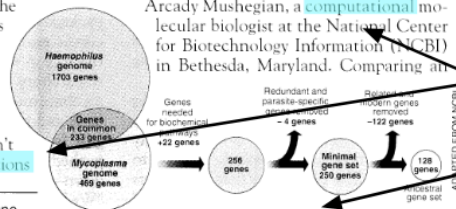
data 0.02
number 0.02
computer 0.01
...

Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK—How many **genes** does an **organism** need to **survive**? Last week at the genome meeting here,* two genome researchers with radically different approaches presented complementary views of the basic genes needed for **life**. One research team, using **computer** analyses to compare known **genomes**, concluded that today's **organisms** can be sustained with just 250 genes, and that the earliest life forms required a mere 128 **genes**. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.

Although the numbers don't match precisely, those **predictions**

"are not all that far apart," especially in comparison to the 75,000 **genes** in the human genome, notes Siv Andersson at Uppsala University in Sweden, who arrived at the 800 number. But coming up with a consensus answer may be more than just a **genetic** numbers game, particularly as more and more **genomes** are completely mapped and sequenced. "It may be a way of organizing any newly **sequenced genome**," explains Arcady Mushegian, a **computational** molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an



* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.

Stripping down. Computer analysis yields an estimate of the minimum modern and ancient genomes.

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Figure 1: **The intuitions behind latent Dirichlet allocation.** We assume that some number of “topics,” which are distributions over words, exist for the whole collection (far left). Each document is assumed to be generated as follows. First choose a distribution over the topics (the histogram at right); then, for each word, choose a topic assignment (the colored coins) and choose the word from the corresponding topic. The topics and topic assignments in this figure are illustrative—they are not fit from real data. See Figure 2 for topics fit from data.

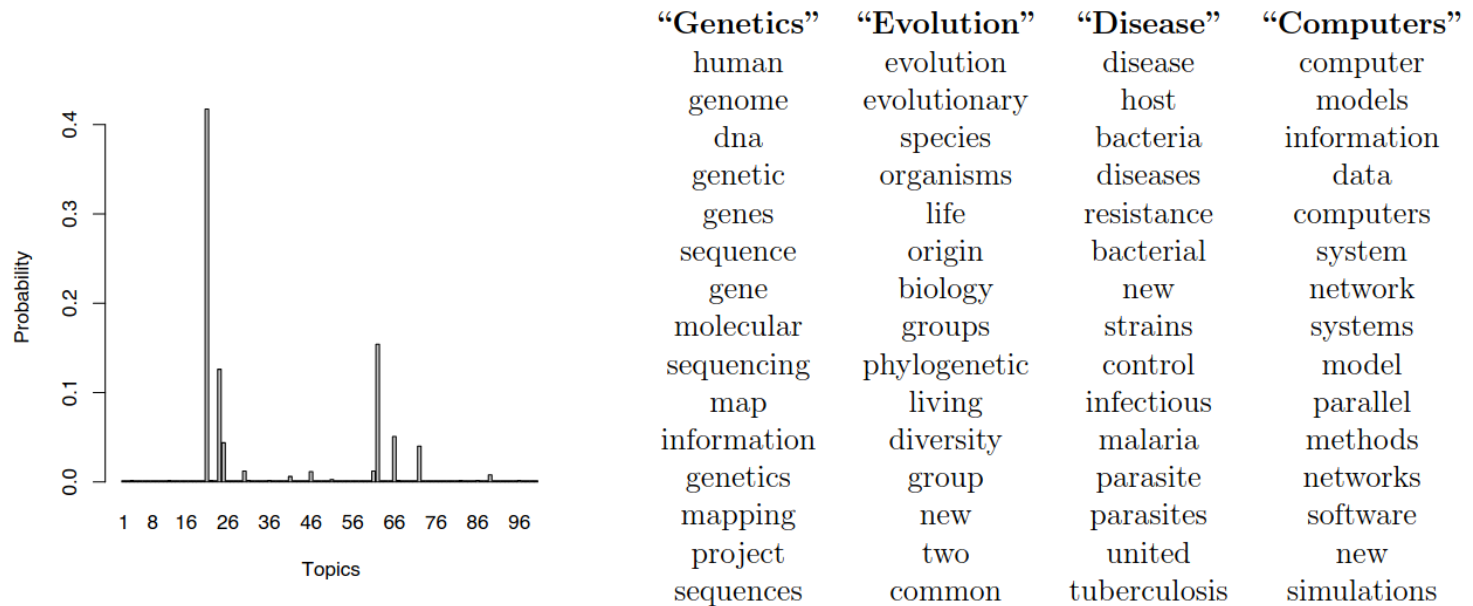


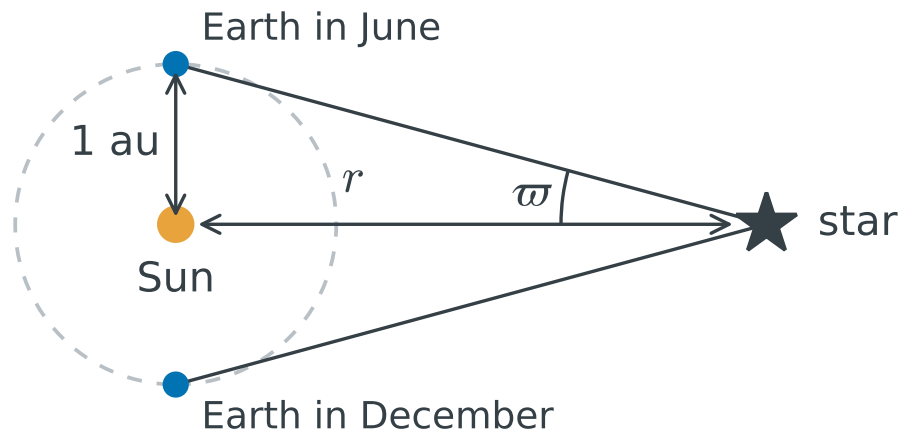
Figure 2: **Real inference with LDA.** We fit a 100-topic LDA model to 17,000 articles from the journal *Science*. At left is the inferred topic proportions for the example article in Figure 1. At right are the top 15 most frequent words from the most frequent topics found in this article.

Example 4: How far are the stars?

Gaia measured the **parallax** of 1.47 billion stars: the angle that the radius of Earth's orbit subtends at the star, which shrinks as the star gets further away,

$$\varpi = \frac{1}{r}.$$

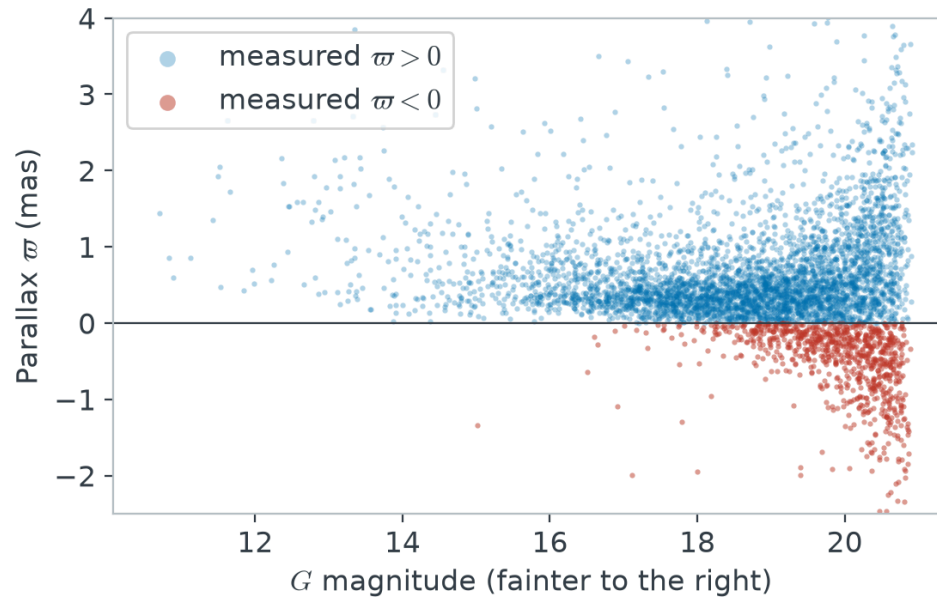
The units make it exact: ϖ in milliarcseconds (a thousandth of an arcsecond) and r in kiloparsecs (about 3260 light-years).



Not to scale.

Distances are what turn a catalogue of angles and brightnesses into physics:

- how bright a star looks depends on how bright it really is and on how far away it is; the distance separates the two, and how bright it really is tells its mass and its age,
- how fast it drifts across the sky, an angle per year, becomes a speed in kilometres per second, which is how the mass of the Galaxy, dark matter included, is weighed,
- the catalogue stops being a picture of the sky and becomes a three-dimensional map.



A random sample of 5000 stars from Gaia DR3. For 82% of them $\varpi_i/\sigma_i < 5$,
and for the 17% below the line $1/\varpi_i$ is not even a distance.

Geometry. The parallax of a star at distance r_i is exactly $1/r_i$. No modelling freedom here: it is what a parallax is.

The instrument. Gaia reports an estimate of that angle, with an uncertainty σ_i computed star by star. Modelling its errors as unbiased and Gaussian gives

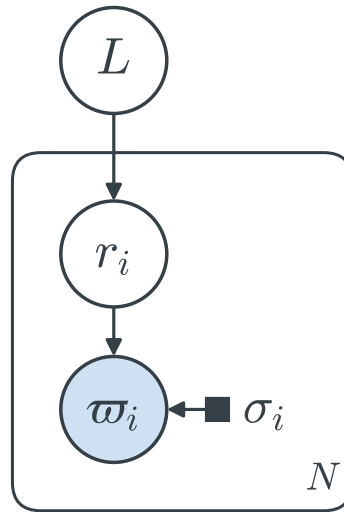
$$p(\varpi_i \mid r_i, \sigma_i) = \mathcal{N}(\varpi_i \mid 1/r_i, \sigma_i^2),$$

which is why a measured parallax can be negative while a distance cannot.

The Galaxy. Stars are not spread evenly: a shell at distance r has a volume growing like r^2 , and their density thins out with distance, modelled as an exponential of scale length L ,

$$p(r_i \mid L) = \frac{r_i^2}{2L^3} \exp(-r_i/L), \quad r_i > 0.$$

That scale length is unknown too, and belongs to the Galaxy rather than to any star, so it gets a vague prior of its own, $p(L) = \text{Uniform}(L \mid 0, 5)$.



Everything unobserved is inferred at once,

$$p(r_{1:N}, L \mid \varpi_{1:N}, \sigma_{1:N}) \propto p(L) \prod_{i=1}^N p(\varpi_i \mid r_i, \sigma_i) p(r_i \mid L),$$

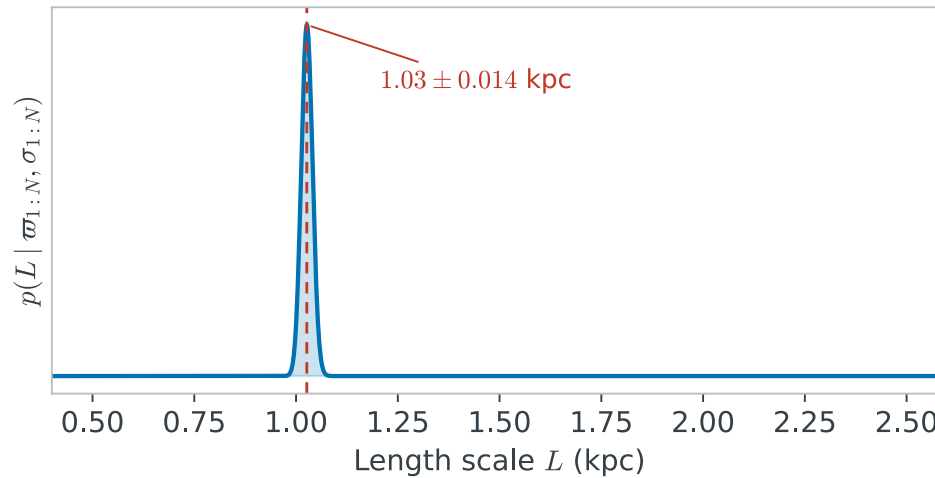
a posterior over $N + 1$ unknowns. Two questions are worth asking of it.

What do the stars say about the Galaxy? Integrating out every distance leaves the length scale alone,

$$p(L \mid \varpi_{1:N}, \sigma_{1:N}) \propto p(L) \prod_{i=1}^N \int p(\varpi_i \mid r_i, \sigma_i) p(r_i \mid L) dr_i,$$

the marginal likelihood of the catalogue, times the prior.

One parallax constrains L almost not at all: a single noisy angle is compatible with nearly any value. Five thousand of them together are not.

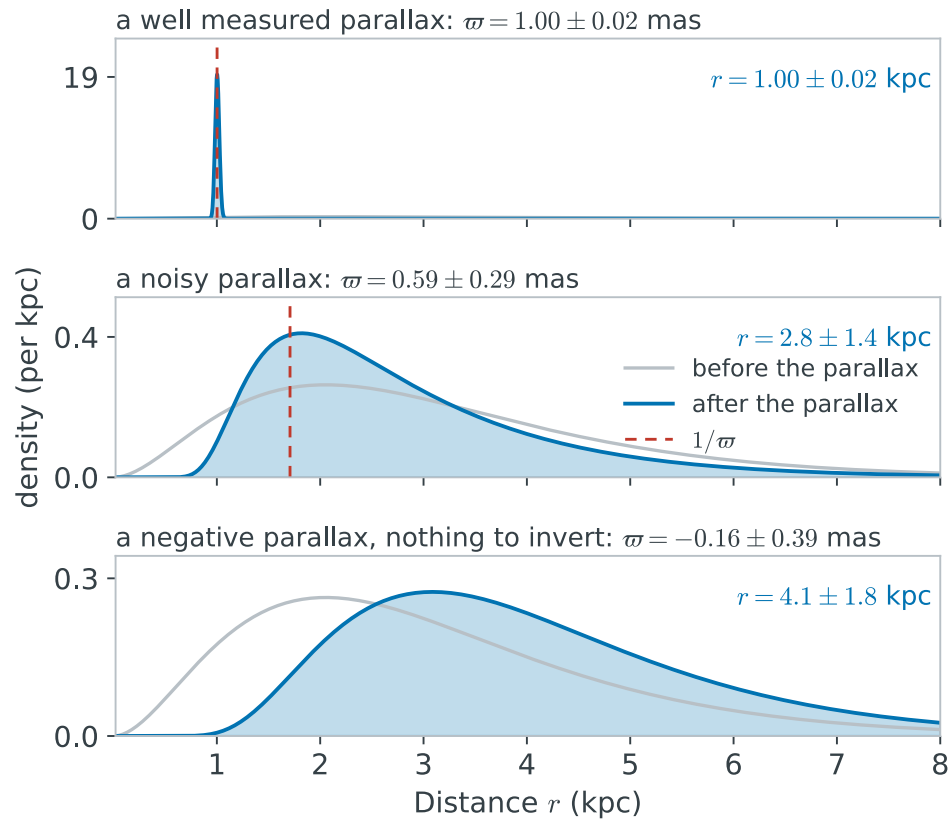


The stellar density thins out along these lines of sight with a scale length of **1.03 ± 0.014** kpc: a measurement of the shape of the Galaxy.

And where is each star? The distance of star i comes from the same posterior, with L and the other distances integrated out,

$$p(r_i \mid \varpi_{1:N}, \sigma_{1:N}) \propto \int p(\varpi_i \mid r_i, \sigma_i) p(r_i \mid L) p(L \mid \varpi_{1:N}, \sigma_{1:N}) dL.$$

When that parallax is precise, it decides the answer on its own. When it is not, the distance comes mostly from what the whole catalogue has established about where stars sit.



Three stars of the sample: how likely each distance is before the parallax is used (grey), and after (blue).



Every term of this model came from somewhere: the geometry from the definition of a parallax, the error model from the instrument, the prior from the way stars fill the Galaxy.

A model is an argument about how the data came to be, not a stack of convenient distributions. Each assumption can be named, defended, and attacked.

Cheat sheet for Gaussian models (Särkkä, 2013)

If $\mathbf{x}$ and $\mathbf{y}$ have the joint Gaussian distribution

$$p \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \mathcal{N} \left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}, \begin{pmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C}^T & \mathbf{B} \end{pmatrix} \right),$$

then the marginal and conditional distributions of $\mathbf{x}$ and $\mathbf{y}$ are given by

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\mathbf{a}, \mathbf{A})$$

$$p(\mathbf{y}) = \mathcal{N}(\mathbf{y}|\mathbf{b}, \mathbf{B})$$

$$p(\mathbf{x}|\mathbf{y}) = \mathcal{N}(\mathbf{x}|\mathbf{a} + \mathbf{CB}^{-1}(\mathbf{y} - \mathbf{b}), \mathbf{A} - \mathbf{CB}^{-1}\mathbf{C}^T)$$

$$p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{y}|\mathbf{b} + \mathbf{C}^T\mathbf{A}^{-1}(\mathbf{x} - \mathbf{a}), \mathbf{B} - \mathbf{C}^T\mathbf{A}^{-1}\mathbf{C}).$$

If the random variables $\mathbf{x}$ and $\mathbf{y}$ have Gaussian probability distributions

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\mathbf{m}, \mathbf{P})$$
$$p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{y}|\mathbf{H}\mathbf{x} + \mathbf{u}, \mathbf{R}),$$

then the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ is Gaussian with

$$p \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \mathcal{N} \left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{m} \\ \mathbf{H}\mathbf{m} + \mathbf{u} \end{pmatrix}, \begin{pmatrix} \mathbf{P} & \mathbf{P}\mathbf{H}^T \\ \mathbf{H}\mathbf{P} & \mathbf{H}\mathbf{P}\mathbf{H}^T + \mathbf{R} \end{pmatrix} \right).$$